

Lecture #14: Probability and Markov Chains

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Definition

The term *experiment* is used to refer to any process whose outcome is not known in advance, such as flipping a coin or rolling a die. The *sample space* associated with an experiment is the set of all possible outcomes. The sample space is usually denoted by Ω , the capital Greek letter Omega

Example

The flip of one coin has two possible outcomes, called “Heads” and “Tails,” and denoted by H and T . Flipping three coins leads to $2^3 = 8$ outcomes:

	HHT	HTT	
HHH	HTH	THT	TTT
	THH	TTH	



Example

The roll of one die has six possible outcomes: 1, 2, 3, 4, 5, 6. Rolling two dice leads to $6^2 = 36$ outcomes $\{(m, n) : 1 \leq m, n \leq 6\}$.

Definition

Intuitively, an *event* is a statement about the outcome of an experiment. Formally, an event is a subset of the sample space.

Example

An example for flipping three coins is “two coins show Heads,” or

$$A = \{HHT, HTH, THH\}$$

An example for rolling two dice is “the sum is 9,” or

$$B = \{(6, 3), (5, 4), (4, 5), (3, 6)\}$$

Definition

Events are just sets, so we can perform the usual operations of set theory on them. For example, if $\Omega = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 3\}$, and $B = \{2, 3, 4, 5\}$, then the *union* $A \cup B = \{1, 2, 3, 4, 5\}$, the *intersection* $A \cap B = \{2, 3\}$, and the *complement of* A , $A^c = \{4, 5, 6\}$. To introduce our next definition, we need one more notion: two events are *disjoint* if their intersection is the empty set, $\emptyset$. A and B are not disjoint, but if $C = \{5, 6\}$, then A and C are disjoint.



Definition

A *probability* is a way of assigning numbers to events that satisfies:

- 1 For any event A , $0 \leq P(A) \leq 1$.
- 2 If Ω is the sample space, then $P(\Omega) = 1$.
- 3 the probability of a union of disjoint events is the sum of the probabilities of the sets.

$$P\left(\bigcup_{i=1}^k A_i\right) = \sum_{i=1}^k P(A_i)$$



Example

For a very general example of a probability, let $\Omega = \{1, 2, \dots, n\}$; let $p_i \geq 0$ with $\sum_i p_i = 1$; and define $P(A) = \sum_i p_i$ for all i in A . Two basic properties that follow immediately from the definition of a probability are

$$P(A) = 1 - P(A^c)$$
$$P(B \cup C) = P(B) + P(C) - P(B \cap C)$$

Example

Roll two dice and suppose for simplicity that they are red and green. Let A = “at least one 4 appears,” B = “a 4 appears on the red die,” and C = “a 4 appears on the green die,” so $A = B \cup C$. Find the Probability of A , $P(A)$ two different ways.

First, A^c = “neither die shows a 4,” which contains $5 \cdot 5 = 25$ outcomes so

$$P(A) = 1 - 25/36 = 11/36.$$

Secondly, $A = B \cup C$, so $P(B) = P(C) = 1/6$ while $P(B \cap C) = P(\{4, 4\}) = 1/36$, so $P(A) = 1/6 + 1/6 - 1/36 = 11/36$.

Conditional Probability

Definition

Suppose we are told that the event A with $P(A) > 0$ occurs. Then the sample space is reduced from Ω to A and the probability that B will occur given that A has occurred is

$$P(B|A) = P(B \cap A)/P(A).$$

To explain this formula, note that only the part of B that lies in A can possibly occur, and since the sample space is now A , we have to divide by $P(A)$ to make $P(A|A) = 1$. Multiplying on each side by $P(A)$:

$$P(B|A)P(A) = P(B \cap A).$$



Conditional Independence Example

Example

Suppose we draw two balls without replacement from an urn with 6 blue balls and 4 red balls. What is the probability we will get two blue balls? Let A = blue on the first draw, and B = blue on the second draw. Clearly, $P(A) = 6/10$. After A occurs, the urn has 5 blue balls and 4 red balls, so $P(B|A) = 5/9$ and it follows that

$$P(A \cap B) = P(A)P(B|A) = \frac{6}{10} \cdot \frac{5}{9}$$

To see that this is the right answer notice that if we draw two balls without replacement and keep track of the order of the draws, then there are $10 \cdot 9$ outcomes, while $6 \cdot 5$ of these result in two blue balls being drawn.



Independence

Definition

Last but far from least, two events A and B are said to be *independent* if $P(B|A) = P(B)$. In words, knowing that A occurs does not change the probability that B occurs. Using the multiplication rule this definition can be written in a more symmetric way as

$$P(A \cap B) = P(A)P(B)$$

Definition

There are two ways of extending the definition of independence to more than two events. $A_1, \dots, A_n$ are said to be *pairwise independent* if for each $i \neq j$, $P(A_i \cap A_j) = P(A_i)P(A_j)$, that is, each pair is independent. $A_1, \dots, A_n$ are said to be *independent* if for any $1 \leq i_1 < i_2 < \dots < i_k \leq n$, we have

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k})$$

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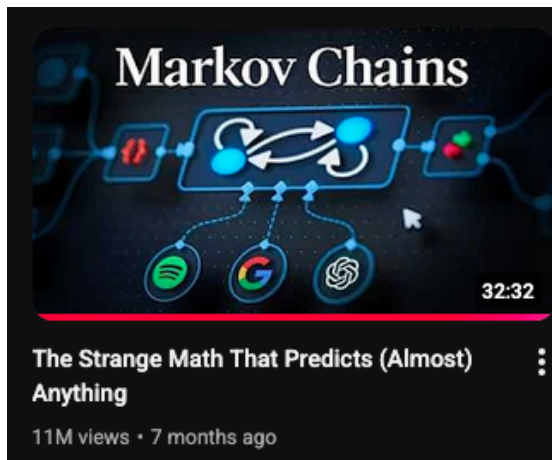
① Probability Overview

② Markov Chains



Markov Chains

The importance of Markov chains comes from two facts: (i) there are a large number of physical, biological, economic, and social phenomena that can be modeled in this way, and (ii) there is a well-developed theory that allows us to do computations.



First Markov Chain

Example

Consider a gambling game in which on any turn you win \$1 with probability $p = 0.4$ or lose \$1 with probability $1 - p = 0.6$. Suppose further that you adopt the rule that you quit playing if your fortune reaches $\$N$. Of course, if your fortune reaches $\$0$ the casino makes you stop.

Definition

Let X_n be the amount of money you have after n plays. Your fortune, X_n has the “Markov property.” In words, this means that given the current state, X_n , any other information about the past is irrelevant for predicting the next state X_{n+1} .

Definition

We say that X_n is a discrete time *Markov chain* with *transition matrix* $p(i, j)$ if

$$P(X_{n+1} = j | X_n = i) = p(i, j)$$

More Analysis of Gambler's Ruin

Describing the Gambler's ruin in terms of a probability matrix.

$$p(0,0) = 1$$

$$p(N,N) = 1$$

$$p(i, i+1) = 0.4, \quad 0 < i < N$$

$$p(i, i-1) = 0.6, \quad 0 < i < N$$

if we make a matrix out of this we have

	0	1	2	3	4	5
0	1.0	0	0	0	0	0
1	0.6	0	0.4	0	0	0
2	0	0.6	0	0.4	0	0
3	0	0	0.6	0	0.4	0
4	0	0	0	0.6	0	0.4
5	0	0	0	0	0	1.0



Graphical View of Markov Chains

We can also view this as a collection of states and ways to move between them.

